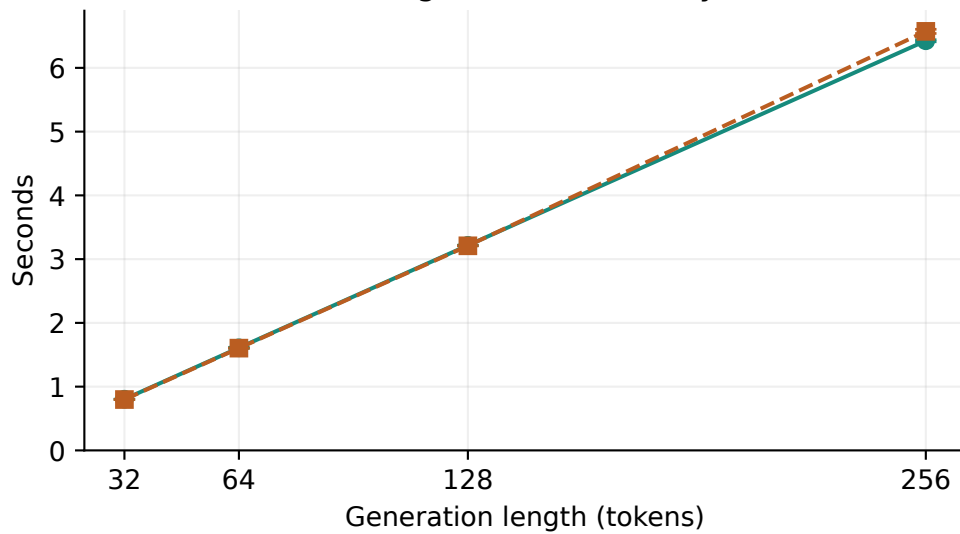


KV Cache ON/OFF: generation length sweep

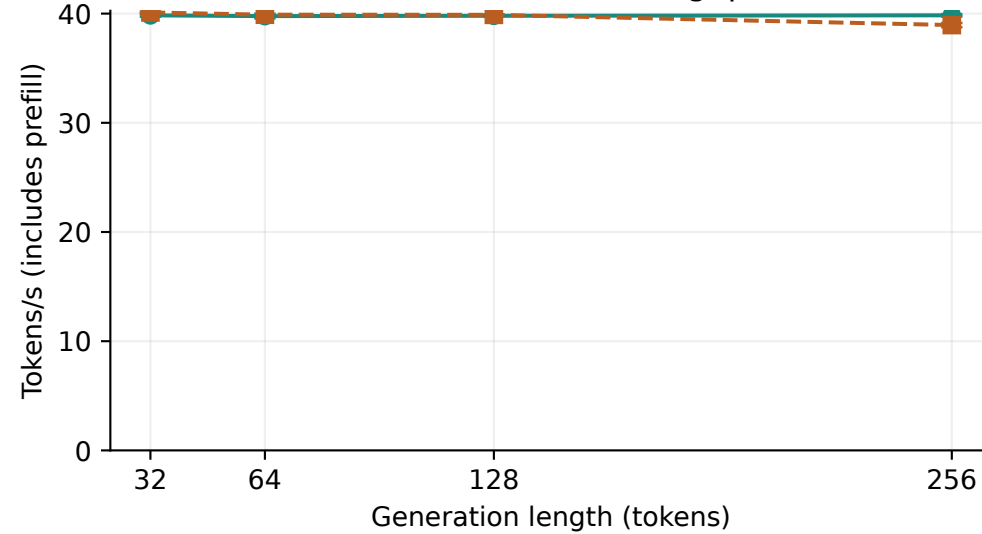
Qwen2.5-1.5B | RTX 3090 | BF16 | eager | prompt 128 | batch 1

Cache ON Cache OFF

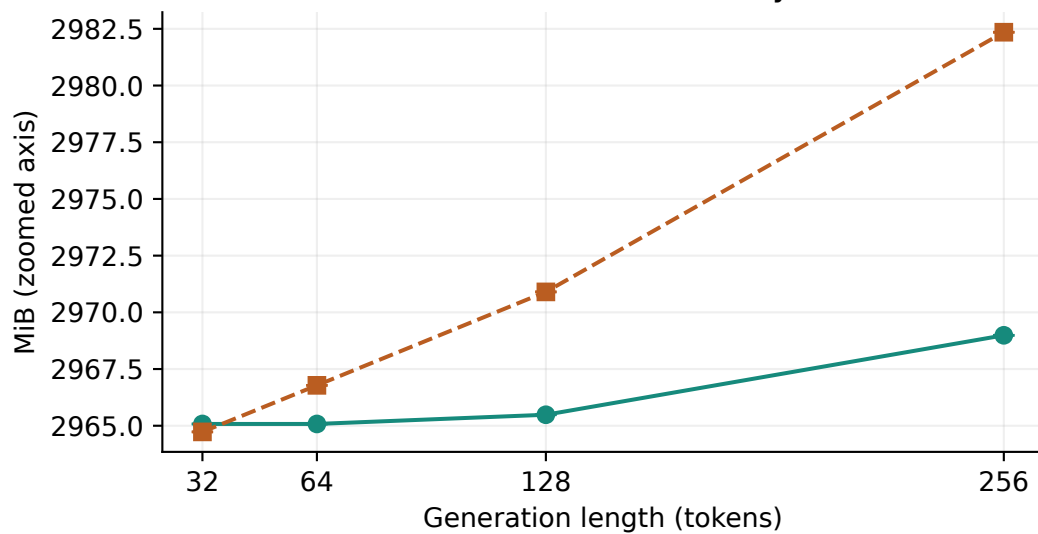
Total generation latency



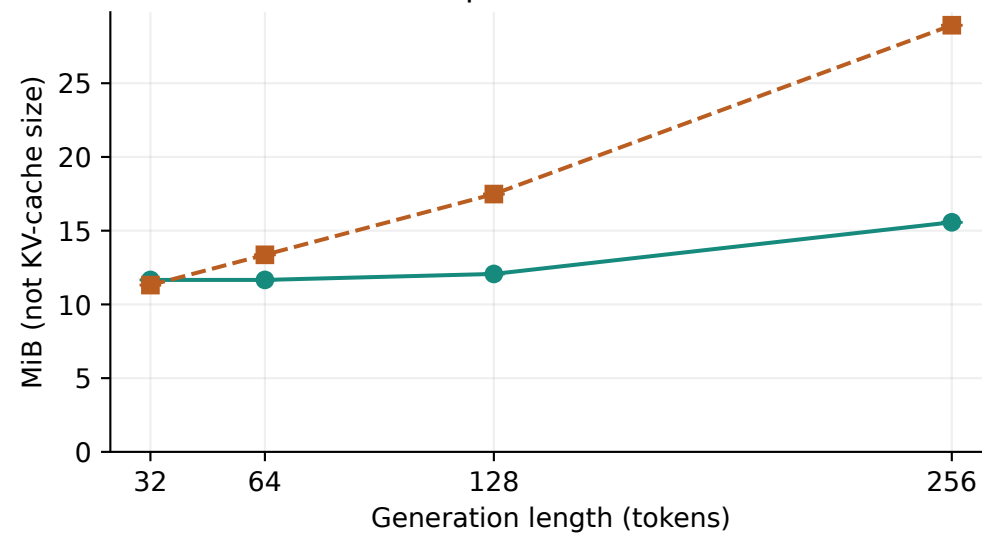
Generated-token throughput



Peak allocated memory



Allocated peak above baseline



Means $\pm$ sample SD (n=5, ddof=1). Warm-up excluded. The original 128 input token IDs are fixed.

Latency speedup (OFF / ON) by generation length: 32: 0.99x; 64: 1.00x; 128: 1.00x; 256: 1.02x

Outputs match across all ten measured runs at each length. Peak memory includes temporary tensors.